

Horizon A1–A6: Runtime Safety with Enforced Singapore Policy Assurance

Tan Wee Joe
University College London
London, United Kingdom
tanweejoe@gmail.com

ABSTRACT

Horizon places independent runtime assurance between a vessel’s decision AI and its actuators. A1–A5 progressively add probabilistic risk, bounded prediction, barrier filtering, and evidence-conditioned recovery. A6 now enforces a bounded Singapore engineering-policy layer inside the actuator gate: normal A5 commands require policy authorization; withholding triggers checked recovery or explicitly unknown minimum-risk action. Historically, A5 achieved 99.83% gate acceptance in 12 development branches. New three-seed normal-transit counterfactuals authorized all 72 normal commands under a permissive test policy; a restrictive speed policy admitted zero normal commands and substituted 72 recovery commands. These are synthetic functional experiments, not operational or legal qualification.

KEYWORDS

runtime assurance, autonomous vessels, maritime policy, mechanistic interpretability

1 PROTECTED SYSTEM ARCHITECTURE

The external decision AI may propose heading and speed but cannot command actuators directly. Each proposal is joined with navigation, contacts, uncertainty, platform status, communications, perception, and neural-health evidence. An assurance candidate produces a decision; an exclusive gate then revalidates identity, lineage, freshness, deadlines, actuator limits, and final-command safety. Unknown, stale, late, or unbounded evidence can only preserve or reduce authority. A watchdog retains minimum-risk authority if supervision stops.

2 A1–A6 PROGRESSION

All six architectures are deterministic engineering implementations; none uses reinforcement learning. A1–A5 are alternative physical-safety candidates sharing one contract. A6 restricts A5’s selected command through the exclusive gate; it does not independently propose maneuvers.

A1 is transparent and fast, but a threshold result does not establish complete trajectory safety. A2 represents uncertainty probabilistically, but its analytic risk score is uncalibrated and is not a hard safety bound. A3 adds finite-horizon plant rollout and demands a complete safe continuation before issuance. A4 searches the nonlinear vessel map for a barrier-compatible correction and validates the result with a full rollout. A5 makes that path evidence-aware: healthy evidence retains nominal bounds and a 6 m/s cap; degraded required evidence expands bounds by 1.5 and caps speed

Table 1: Architectural progression and present disposition.

ID	Primary mechanism	Disposition
A1	CPA/TCPA thresholds and Simplex switching	baseline
A2	Analytic Gaussian collision-risk approximation	research
A3	Bounded rollout plus complete recovery continuation	retained
A4	Robust barrier correction plus 3DOF validation	retained
A5	Health-conditioned bounds, prediction, A4 filter, recovery	control lead
A6	Gate-enforced Singapore policy checks on exact A5 output	policy layer

at 3 m/s; invalid required evidence forces fallback. It attempts predictive recoverability, invokes A4 when needed, and falls back on incomplete, infeasible, or late results.

A6 evaluates bounded engineering proxies for Rules 5–8 and 13–17. The development profile is a 12 m, 40 GT Singapore government USV under remote supervision in daylight and clear visibility. Source artifacts are hash-checked at load; fresh evidence binds the snapshot and command. Actual course/speed changes override adapter action claims. Unknown, unsupported, late or unsatisfied evidence withholds authority. Normal dispatch must precede policy, evidence and cycle expiry. Emergency recovery remains available and is labelled as a policy override. MASS design references and the broader maritime-policy context motivate the bounded model [1–3]; these engineering predicates do not establish statutory compliance.

3 A1–A5 COMPARATIVE RESULTS

The initial development matrix contained 12 paired identities: four scenarios, three seeds, and five candidates per identity, for 60 branches. Scenarios covered recoverable crossing, dense traffic, vessel degradation, and initially unrecoverable encounters. Timing was idealized except for 20 ms candidate and gate service, so the study compares architecture behaviour rather than production latency.

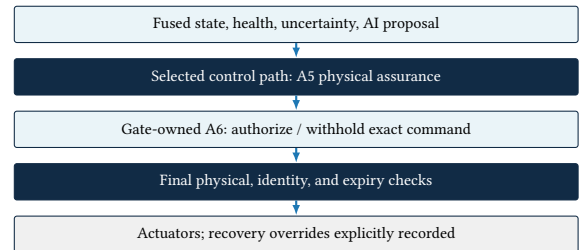


Figure 1: Normal actuation requires physical and policy authorization.

Table 2: A1–A5 development study; 12 branches per candidate.

ID	Gate accepted	Deadline misses	Recoverable collisions	Unsafe/stale accepted
A1	8.87%	0	0	0
A2	8.73%	1	0	0
A3	99.66%	6	0	0
A4	99.44%	10	0	0
A5	99.83%	3	0	0

Table 3: New normal-transit counterfactuals: 3 seeds, 5 s each.

Measure	Result
A5-only normal commands accepted	72
A6 permissive: authorized / normal accepted	72 / 72
A6 speed veto: normal / recovery accepted	0 / 72
A6 speed veto: initial vetoes / latched recoveries	3 / 69
Command-trace mismatches / collisions	0 / 0

None of the 45 declared-recoverable branches collided; all 15 deliberately unrecoverable branches collided. No unsafe or stale result was accepted. A1 and A2 often passed proposals the independent predictive gate rejected, producing acceptance below 9%. A3–A5 actively selected recovery or minimum-risk commands and exceeded 99.4% acceptance. A5 had the highest acceptance and fewer deadline misses than A3 or A4, so it became the physical-control lead. These small, development-set results do not prove statistical superiority or operational safety.

4 A6 SHADOW BASELINE AND ENFORCEMENT

The earlier shadow run reused the 12 historical A5 episode identities, hashes, policies and 30-second horizon. It observed 1,788 decisions: 888 recovery, 885 minimum-risk and 15 invalid. All withheld policy support; median A6 latency was 0.178 ms. This established conservative shadow behavior, but could not demonstrate enforcement on permissive commands.

The enforcement experiment compares A5 alone with two synthetic policy bundles on the same normal-transit seeds. The 6 m/s test cap permits normal commands; the 1 m/s cap vetoes the first command in each seed and latches recovery thereafter. Each arm also records 69 watchdog commands and 72 command-expiry events. The local-acceptance timing model has zero simulated gate service; measured wall work is separate. These results establish functional authorization and veto behavior, not timing readiness or uninterrupted autonomy. The implementation also corrects recovery priming’s simulated-to-host deadline mapping.

The original 12-episode, 30-second matrix was also rerun with enforcement: 1,770 gate acceptances and 18 deadline rejections, comprising 885 recovery, 885 minimum-risk and 18 invalid decisions. No normal command reached policy evaluation; these actions were explicit emergency overrides. Trace mismatches were zero; the three initially-unrecoverable episodes remained unsafe. Host-sensitive deadline counts are revision/run-specific.

5 WHY THE A5+A6 COMPOSITION WAS SELECTED

A5 remains the physical-control lead: it combines evidence-conditioned conservatism, recovery validation and favorable historical acceptance. **A6 is chosen as mandatory policy authorization:** the new counterfactuals show that changing policy changes which physically permissible commands reach the plant. Its assessments bind the complete input, decision, evidence and versioned bundle. This closes the previous observation-only gap while retaining physical revalidation and explicit emergency authority.

Restricted visibility, channel/TSS navigation, unknown vessel classes and broader profiles withhold authorization. Weapons, targeting, classified doctrine and ROE remain excluded. This release vetoes A5’s single selected command; it does not search alternative policy-compliant maneuvers. Recorded operational sources, live evidence integration, legal review and target-platform timing remain qualification work.

6 NEURAL-HEALTH INPUT AND VERDICT

H5 monitors WaSR-T’s temporal-fusion representation with a hash-pinned sparse feature. Its selected development feature beat random-direction and equal-norm causal controls in all 18 trials. An optional simulation-warning mode can reduce the AI’s proposed speed; operational calibration remains unqualified. H5 cannot prove obstacle-free water or increase A5/A6 authority.

The product now defaults to *A5 with enforced A6 authorization*. Without configured recorded sources and fresh policy evidence, normal autonomy is withheld. Recovery overrides are visible in gate telemetry and receipts. The delivered software enforces the bounded policy model; the evidence does not qualify autonomous vessel deployment or full maritime-law compliance.

Reproducibility. Counterfactuals: `scripts/benchmark_a6_enforcement.py`. Result: `horizon-runs/development/a6-enforcement-counterfactual-20260925-v5/summary.json`. SHA-256: `0f63c151344bf6f576257d38efc0302c511f0df127987d9f95223dbbf32543f1`. Original scenario matrix: `experiment/manifests/a6-a32-enforcement-development.json`.

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